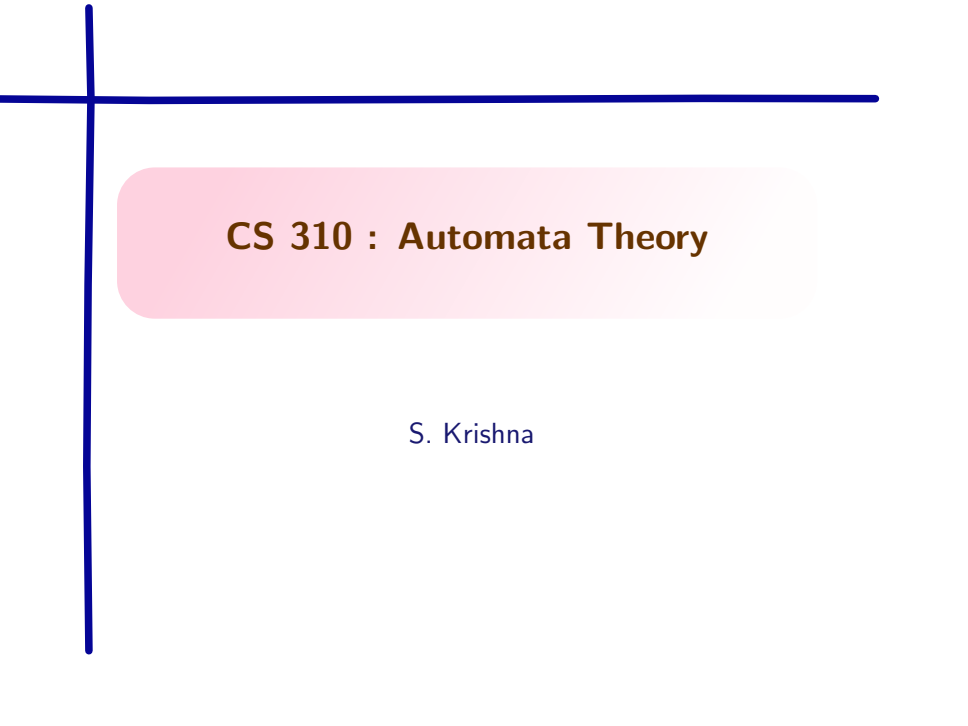


A decorative blue crosshair consisting of a vertical line and a horizontal line intersecting at the top-left corner of the slide.

CS 310 : Automata Theory

S. Krishna

Recap : Acceptance conditions

Theorem

Let $P = (Q, \Sigma, \Gamma, \delta, q_0, \perp, F)$ be a NPDA.

1. If P accepts by final state, there exists NPDA P' accepting by empty stack such that $L(P) = L(P')$

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Languages of NPDA

Context-free Languages (CFLs)

A language $L \subseteq \Sigma^*$ is called **context-free** if there exists a NPDA A such that $L = L(A)$.

Deterministic PDA (DPDA)

Definition

A PDA $A = (Q, \Sigma, \Gamma, \delta, q_0, \perp, F)$ is **deterministic** if for each $q \in Q$ and $X \in \Gamma$

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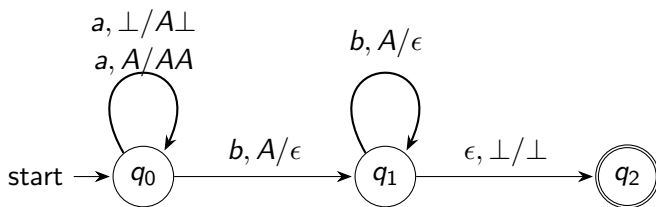
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Examples

1. $\{a^n b^n \mid n \geq 1\}$
2. $\{wcw^R \in \{0, 1, c\}^* \mid w \in \{0, 1\}^*\}$
3. Any regular language

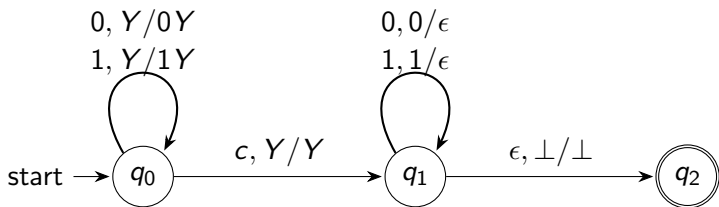
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DPDA are strictly less powerful than PDAs.

The following language cannot be recognized by any DPDA.

$$L_{ww^R} = \{ww^R \mid w \in \{0, 1\}^*\}$$

No DPDA for $\{ww^R \mid w \in \{0,1\}^*\}$

Claim

$L = \{ww^R \mid w \in \{0,1\}^*\}$ is context-free, but no DPDA accepts it.

- Intuition : do we know when to start matching?

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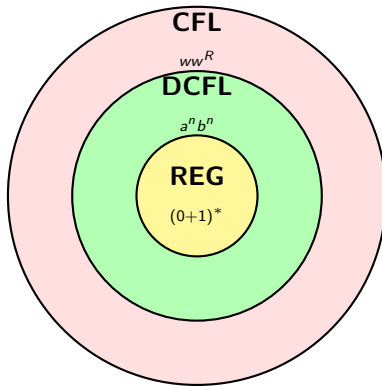
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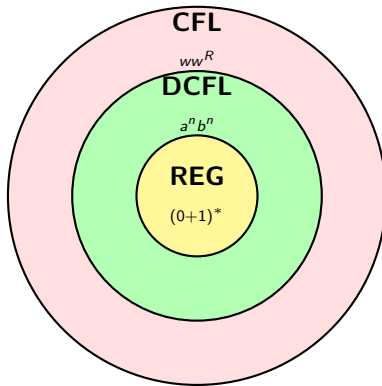
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- ▶ Now consider $w' = (0^n 1 10^n)(0^m 1 10^m)$. Must be accepted only when $m = n$. But the stack will already have no useful information at the end of w , to process the rest of the word and verify $m = n$!

Language hierarchy: $\text{REG} \subsetneq \text{DCFL} \subsetneq \text{CFL}$

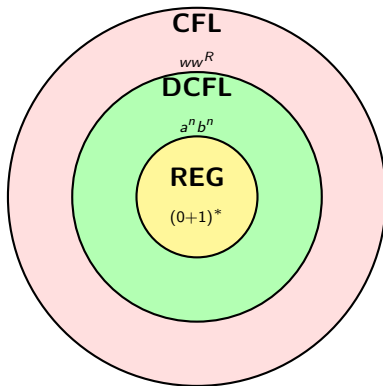


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What if a DPDA instead accepts by **empty stack**? Does final state and empty stack acceptance coincide for DPDAs?

Do we still get $\text{REG} \subsetneq \text{DCFL} \subsetneq \text{CFL}$?

DCFLs

Adding back what I said in class

- ▶ Prefix-free languages : those where if $w \in L$, then no proper prefix of w is accepted
- ▶ Can you construct a DPDA accepting by empty stack for $\{0, 1\}^*$?
- ▶ Is $\{0, 1\}^*$ prefix free?

Properties of CFLs

- ▶ Are they closed under union?
- ▶ Are they closed under intersection?
- ▶ Are they closed under complementation?

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- ▶ Recall: for regular languages we had regular expressions to write rules?
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